C. Chocolate Bunny

time limit per test: 1 second
memory limit per test: 256 megabytes

This is an interactive problem.

We hid from you a permutation p of length n, consisting of the elements from 1 to n. You want to guess it. To do that, you can give us 2 different indices i and j, and we will reply with $p_i \bmod p_j$ (remainder of division p_i by p_j).

We have enough patience to answer at most $2 \cdot n$ queries, so you should fit in this constraint. Can you do it?

As a reminder, a permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array) and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Input

The only line of the input contains a single integer n ($1 \le n \le 10^4$) — length of the permutation.

Interaction

The interaction starts with reading n.

Then you are allowed to make at most $2 \cdot n$ queries in the following way:

- "`? x y`" ($1 \le x, y \le n, x \ne y$).

After each one, you should read an integer k, that equals $p_x \bmod p_y$.

When you have guessed the permutation, print a single line "`! `" (without quotes), followed by array p and quit.

After printing a query do not forget to output end of line and flush the output. Otherwise, you will get `Idleness limit exceeded`. To do this, use:

- `fflush(stdout)` or `cout.flush()` in C++;
- `System.out.flush()` in Java;
- `flush(output)` in Pascal;
- `stdout.flush()` in Python;
- see documentation for other languages.

Exit immediately after receiving "`-1`" and you will see `Wrong answer` verdict. Otherwise you can get an arbitrary verdict because your solution will continue to read from a closed stream.

Hack format

In the first line output n ($1 \le n \le 10^4$). In the second line print the permutation of n integers $p_1, p_2, \ldots, p_n$.

Example

input

```
3
1
2
1
0
```

output

```
? 1 2
? 3 2
? 1 3
? 2 1
! 1 3 2
```

Codeforces Round 669 (Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++20 13.2 (64 bit, win

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
326594223	Jun/30/2025 05:59	Accepted
326594078	Jun/30/2025 05:58	Accepted

→ Problem tags

constructive algorithms, interactive, math, two pointers, *1600

No tag edit access

→ Contest materials

- Announcement
- Tutorial